

Online appendix I: Pre-analysis plan

Resource Rents and Deliberative Democracy in Tanzania

*Pre-analysis Plan*¹

Nancy Birdsall, James Fishkin, Mujibu Moyo, Justin Sandefur²

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Abstract

In 2010, Tanzania discovered natural gas reserves off its southern coast worth roughly fifteen times its annual GDP at prevailing prices. The large literature on the resource curse suggests this new gas discovery brings considerable risks to a low-income country with already weak governance institutions. How should Tanzania manage this new revenue to avoid the traditional resource curse? We will report on multiple rounds of a nationally representative opinion poll soliciting ordinary Tanzanians' views on how to manage and spend gas revenues. Between polling rounds, a random subset of poll respondents will receive detailed information about the natural gas discovery, and the pros and cons of various gas policy options. A randomly drawn subset of individuals receiving the information treatment will be invited to a national deliberative event, where they will debate gas policy options and participate in question and answer sessions with experts. We will report treatment effects of information and deliberation on support for alternative uses of gas revenue, including saving versus spending, and direct distribution of resource rents. We will also test a number of hypotheses about the process of deliberation itself, related to peer effects, leader effects, and polarization.

1 Summary of experimental design & interventions

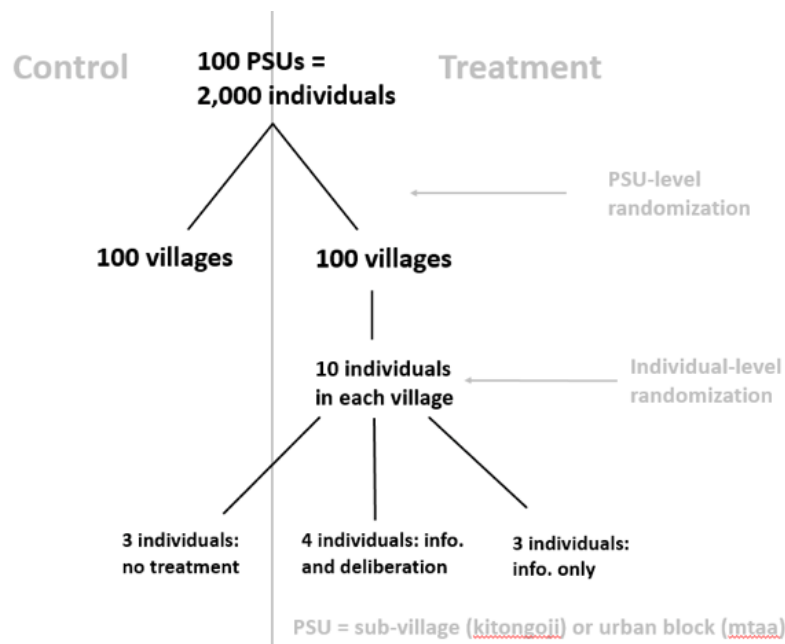
A nationally representative baseline poll conducted between February and April, 2015 asked a target of 2,001 adult Tanzanians in 200 rural and urban clusters about their knowledge and policy preferences regarding Tanzania's recent natural gas discovery and the use of any future revenues. All respondents received a mobile phone to facilitate follow-up polling by phone.

In each of 100 randomly selected survey clusters, 7 of 10 respondents were invited to view a documentary video presenting pros and cons of various gas policy options (the information treatment). Of the respondents who were invited to the information treatment, 4 individuals were invited to a democratic deliberation about the use of the natural gas.

The basic design is presented in the figure below. (The graph uses “villages” and PSUs interchangeably, though some PSUs are in fact urban.)

¹ Note that this draft of the analysis plan was completed and registered after baseline data collection was completed, but before the main intervention described here (i.e., the deliberative event) and before any follow-up data has been collected. Further details of this experiment are available at the American Economic Association RCT registry at <http://www.socialscisearch.org/trials/689>.

² Bobby Fishkin, Faraz Haqqi, Jen Richmond, and Alice Siu have contributed enormously to the design of this project.



The 400 invitees to the deliberative event will be divided into 25 randomly assigned small groups to discuss a sequence of gas policy options, followed by plenary Q and A sessions with a panel of policymakers and experts in the natural gas sector. The discussions will be broken into four rounds over two days, and groups will be randomly reassigned after the first day.

Follow-up polling will measure the medium-term impact (i.e., weeks rather than hours after treatment) of information and deliberation on respondents' knowledge and policy preferences. Follow-up polling will be conducted by phone and broken into multiple rounds, due to limitations on respondents' attention span on the phone. See the entry for this study in the AEA RCT registry for further details.

2 Hypotheses about natural gas policy preferences and knowledge

Data analysis of survey results will test a total of six substantive hypotheses about the effect of information and deliberation treatments on natural gas policy preferences and knowledge.

In each case we test the null hypothesis that each treatment will have no impact on the following outcomes.

H1: Support for **extracting** and exporting natural gas.

H2: Support for **saving** rather than spending gas revenue.

H3: Support for government spending versus **direct distribution** of rents.

H4: Preference for government spending on **social services** versus infrastructure, transport, and industry.

H5: Support for **transparency** and oversight of gas revenues,

H6: **Knowledge** of the natural gas discovery

Each outcome will be measured by the mean of the standardized responses to several survey questions. Full details are found in the hypothesis table below.

We will also test heterogeneous effects for each of these hypotheses along four dimensions: education and knowledge; wealth; trust in government; and gender. The components of the first three measures of heterogeneity are also described in the table at the end. For education and knowledge, the mean effects

index will combine just two components – the total number of correct answers on the knowledge and literacy test, and years of schooling. For wealth, we use a predicted consumption measure, based on the coefficients on a regression of household consumption on survey items in National Panel Survey that overlap with our questionnaire (these items were included in the questionnaire for this purpose). For trust in government, we propose to use a simple mean effects index of the items listed in the table below.

3 Hypotheses about the process of deliberation

In this section we specify three hypotheses related to the process of deliberation itself. The analysis will exploit the random assignment of participants to 25 deliberative groups with a target size of 16 people each (we anticipate some drop-off between the baseline and deliberation).

Note that groups will be reassigned between day 1 and day 2 of deliberations, creating a total of 50 exogenous groupings. The topic of deliberation will differ between days 1 and 2, and deliberation topics correspond to different sets of outcome variables. We anticipate the strongest effect of group assignment to be on questions related to the topics covered by that group that day, but will also test whether day 1 groups affect responses on questions related to day 2 topics and vice versa.

H7. Peer effects: Individual responses will be (unequally) influenced by the views of other participants in the deliberation..

We expect social interactions to be a key component of deliberation. Peers, by which we mean members of the same small group in deliberation, will influence each other. But a large literature on, e.g., jury deliberations implies that certain disadvantaged groups may be drowned out. Also related to the role of peer effects, [Sunstein \(1999\)](#) predicts that deliberation will “move groups, and the individuals who compose them, toward a more extreme point in the direction indicated by their own pre deliberation judgments.” To test these broad theories, we hypothesize:

- 7a. Group-level polarization: Participants’ views will change in the direction of their peers' average views at baseline. Thus deliberation will increase the strength of majorities within deliberative groups.
- 7b. Individual-level polarization: Deliberation will move participants’ views away from neutrality, toward the extremes relative to their pre-deliberation responses.
- 7c. Members of disadvantaged groups (women, the uneducated, the poor) will participate less in deliberation, as measured by frequency of comments. [We will test this hypothesis only if data on participation is successfully collected for at least 80% of groups – there is some uncertainty about the available technology to capture this information in our venue.]
- 7d. Members of disadvantaged groups will exert less influence on the post-deliberation opinions of their fellow group members, i.e., the coefficient on average baseline views of female peers will be smaller than that of male peers.

H8. Leader effects: Leaders' views will determine the outcome of deliberation.

We will test for "leader effects" ([Humphreys, Masters, and Sandbu, 2006](#)) by regressing respondent preferences post-deliberation on baseline preferences of the moderator in each respondent's randomly assigned deliberation group together with our standard controls.

NB: Because moderators only moderate a single session on a given topic, we cannot measure leader effects through a leader fixed effects regression a la Humphreys et al.

We will also test three mechanisms underlying leader effects described in Humphreys et al., which we specify as sub-hypotheses:

- 8a. Suasion: Participants' poll response will move in the direction of leaders' views.
- 8b. Self-censorship: Opinions expressed during deliberation will conform to leaders' views.
- 8c. False relay: Leaders will exaggerate the extent to which deliberation conformed to their own views.

To test these mechanisms, we will measure leader effects on three outcome indicators: (i) participants' post-deliberation poll responses, (ii) leaders' assessment of the outcome of deliberation, and (iii) a separate assessment of the outcome of deliberation by an independent observer. Hypothesis 1a can be measured by impacts on (i). Hypothesis 1b implies that leaders' effect on (iii) will be greater than their effect on (ii). Hypothesis 1c implies leader effects on (iii) will be smaller than their effects on (ii).

4 Data Collection

Data will be collected at two main stages: the baseline and follow-up polls, with the later broken into two rounds of (mostly) non-overlapping questions.

- Baseline survey. This survey will be conducted face-to-face in randomly-sampled respondents' houses throughout the country. We will ask a series of socioeconomic questions based on questions asked in Tanzanian versions of the Afrobarometer and CWIQ surveys. These questions will be divided into modules that address the following: demographic details, education level, employment status, ownership of assets and amenities, governance, public engagement, and trust in institutions. In addition to these questions, we have created a new module of natural resource questions regarding Tanzania's impending natural gas revenues.
- Follow-up mobile phone surveys. We will leave out the socioeconomic questions during follow-up surveys since the outcomes we want to measure concern the natural resource questions. Therefore, we will instead ask only the questions in the natural resources module.

In addition, auxiliary data collection will include: (i) subjective assessment of the outcome of deliberation by moderators ("leaders") and note takers; and (ii) tabulations of the frequency of participation in deliberation by each participant.

5 Econometric specification and data analysis

Intent-to-treat analysis: All treatment effects will be calculated on the basis of intention to treat, i.e., examining the impact of random assignment, regardless of compliance with that assignment. If compliance fall below 80% for any treatment, we will also report average treatment effects on the treated, using random assignment as an instrumental variable.

Specification: We will estimate the impact of treatments on follow-up poll responses in an analysis of covariance (ANCOVA) specification, regressing post-treatment responses on pre-treatment responses and the treatment dummy ([McKenzie 2012](#)). We will report results both with and without additional demographic and geographic controls (gender, age, district dummies).

Multiple comparisons: The main hypothesis table lists six main hypotheses. Each of these consists of several survey questions. We will measure treatment effects on a mean effects index ([Kling et al 2007](#)) for each of these main hypotheses. We will also report results for each of the constituent variables in the indices, with p-values corrected for multiple comparison.

Attrition: We will report [Lee \(2009\)](#) bounds on any statistically significant treatment effects on our main hypotheses.

Null hypotheses: Treatment will not affect...		Indicators	Variable name(s)
H1	Support for <u>extracting and exporting</u> natural gas	Support for extracting gas	No baseline
		Support for using gas for domestic energy consumption	No baseline
		Support for using gas to subsidize domestic energy versus exporting to earn government revenue	No baseline
		Support for exporting gas and paying the full price for gas and using earnings from selling gas to spend on public goods	s2q5
		Support for exporting the gas at the highest possible price	s2q6
H2	Support for <u>saving rather than spending</u> gas revenue.	Support for strict limits on spending	s2q3
		Aversion to spending money on building things for the people, such as roads and the electricity system	s2q7
		Aversion to spending money on public services, such as health care and education	s2q8
		Save money for future generations	s2q11
		Support for saving the majority of money for the future as opposed to spending more money now	s2q20
		Support for saving as opposed to using expected money from gas to borrow money and spend on domestic needs sooner	s2q27
		Belief that when government increases spending on schools & roads a lot of money is wasted.	s2q26
H3	Support for government spending versus <u>direct distribution</u> of rents.	Give money directly to households	s2q9
		Give money to households with children or the elderly	s2q10
		Giving money to households will fight poverty and hunger	s2q14
		Giving money directly to people will help children in poor families have better nutrition and a greater chance of doing well in school	s2q15
		Giving money directly to households will make the government more accountable for the amount people expect each year.	s2q16
		Giving money directly to people as opposed to giving it to government to spend through the national budget	s2q22
		Support for a child wealth account	s2q24
		Support for an adult wealth account	s2q25

Null hypotheses: Treatment will not affect...		Indicators	Variable name(s)
H4	Support for government <u>spending on social services</u> versus infrastructure, transport, and industry	Using the money to build public goods will create jobs	s2q12
		Using the money to build public goods will help the economy grow faster	s2q13
		Spending money on social services will help poor people	s2q17
		Spending money on social services will help the economy grow faster	s2q18
		Building public goods as opposed to providing social services	s2q21
H5	Support for <u>transparency and oversight</u> of gas revenues, or additional restrictions on their use.	Support for publishing oil and gas contracts	s2q2
		Support for money being managed by an independent and international group of experts	s2q4
		Belief that government will spend money on that which is most needed	s2q19
H6	<u>Knowledge</u> of the natural gas discovery.	Knowledge of Tanzania's natural gas discoveries	s2q1
		Where the natural gas was discovered	s2q28
		Knowledge of under whose presidency recent offshore natural gas discoveries were made	s2q29
		Knowledge of whether the government has started receiving revenues from natural gas	s2q30
		Knowledge of expected revenue amount compared to the cost of a new hospital	s2q31a
		Knowledge of expected revenue amount compared to the cost of 10 new regional hospitals	s2q31b
		Knowledge of expected revenue amount compared to the entire national government budget	s2q31c
		Knowledge of expected revenue amount compared to the size of the whole national economy	s2q31d
		Knowledge of expected annual revenue divided equally among Tanzanian citizens	s2q32
Covariate index		Indicators	Variable name(s)
X1	<u>Education</u> and general knowledge	Knowledge of current Prime Minister	s3q16
		Knowledge of current Minister of Finance	s3q17
		Knowledge of current member of parliament	s3q18
		Knowledge of next national election year	s3q19y
		Knowledge of next national election month	s3q19m
		Knowledge of the total population of Tanzania	s3q20

Null hypotheses: Treatment will not affect...		Indicators	Variable name(s)
		Knowledge the president's commission to review the Constitution	s3q21
		Knowledge of which constitution will guide the upcoming national elections	s3q22
		Knowledge of Tanzanian industry that has been criticized in the past for providing favorable contracts to companies	s3q23
		Knowledge of number of regions in Tanzania	s3q24
		Knowledge of which lake forms the border between Tanzania and Malawi	s3q25
		Knowledge of the sector in which most Tanzanian citizens work	s3q26
		Ability to read and write	s4q1
		Highest grade completed	s4q2
X2	<u>Wealth,</u> exposure to markets	Employment in past seven days	s5q1
		Economic activity on which you spend most of your time	s5q2
		Household ownership of telephone	s5q3a1
		Household ownership of stove	s5q3a2
		Household ownership of livestock	s5q3a3
		Household ownership of plough	s5q3a4
		Household ownership of radio and radio cassettes	s5q3a5
		Household ownership of hoes	s5q3a6
		Number of telephones	S5q3b1
		Number of pots	s5q3b2
		Number of livestock	s5q3b3
		Number of ploughs	s5q3b4
		Number of radios and radio cassettes	s5q3b5
		Number of household hoes	s5q3b6
		Type of toilet used by the household	s5q4
		Household roofing material	s5q5
		Main fuel used for cooking	s5q6
		Main fuel used for lighting	s5q7
		Consumption of meat, fish, and animal products	s5q8a
		Consumption of milk/milk products	s5q8b
		Consumption of sugar/sugar products/honey	s5q8c
		Consumption of fruits	s5q8d
		Food insecurity	s5q9
X3	<u>Trust in government,</u> satisfaction with public services,	Belief that leaders at national level represents citizens' interests	s3q1
		Belief that government at the local level represents citizens' interests	s3q1b

Null hypotheses: Treatment will not affect...		Indicators	Variable name(s)
	authoritarian leaning	Belief that leaders at the national level are able to provide solutions to citizens' problems	s3q2
		Belief that leaders at the local level are able to provide solutions to citizens' problems	s3q2b
		Belief that government at the national level spend money responsibly	s3q3
		Belief that leaders at the local level spend money responsibly	s3q3b
		Trust in national government	s3q4
		Trust in the President	s3q7a
		Trust in parliament	s3q7b
		Trust in your local government council	s3q7c
		Trust in the ruling party	s3q7d
		Trust in opposition political parties	s3q7e
		Trust in the police	s3q7f
		Trust in courts of law	s3q7g
		Perception of corruption among the President and Officials in his Office	s3q8a
		Perception of corruption among members of parliament	s3q8b
		Perception of corruption among your local government council	s3q8c
		Perception of corruption among the ruling party	s3q8d
		Perception of corruption among opposition political parties	s3q8e
		Perception of corruption among the police	s3q8f
		Perception of corruption in the courts of law	s3q8g
		Perception of municipal/local government's ability to maintain local roads	s3q9a
		Perception of municipal/local government's ability to maintain local market places	s3q9b
		Perception of municipal/local government's ability to maintain local schools	s3q9c
		Perception of municipal/local government's ability to maintain health clinics	s3q9d
		Perception of municipal/local government's ability to manage the use of the land	s3q9e
		Perception of ease to find out how government uses revenues from people's taxes and fees	s3q10
		Satisfaction with MP's performance	s3q11
		Vote share for the ruling party	s3q15